

Communicating clinical research to reduce cancer risk through diet: Walnuts as a case example



Inflammation is one mechanism through which cancer is initiated and progresses, and is implicated in the etiology of other conditions that affect cancer risk and prognosis, such as type 2 diabetes, cardiovascular disease, and visceral obesity. Emerging human evidence, primarily epidemiological, suggests that walnuts impact risk of these chronic diseases via inflammation. The published literature documents associations between walnut consumption and reduced risk of cancer, and mortality from cancer, diabetes, and cardiovascular disease, particularly within the context of the Mediterranean Diet. While encouraging, follow-up in human intervention trials is needed to better elucidate any potential cancer prevention effect of walnuts, per se. In humans, the far-reaching positive effects of a plant-based diet that includes walnuts may be the most critical message for the public. Indeed, appropriate translation of nutrition research is essential for facilitating healthful consumer dietary behavior. This paper will explore the translation and application of human evidence regarding connections with cancer and biomarkers of inflammation to the development of dietary guidance for the public and individualized dietary advice. Strategies for encouraging dietary patterns that may reduce cancer risk will be explored.